

SRS

Software Requirements Specification (SRS)

Project: Endur – Performance Review & Feedback Management System

Team: 39_endur

Version: 1.03

1. Preface

1.1 Expected Readership

This document is intended for:

- **Project Developers (Team 39_endur)** – Technical understanding and implementation
- **Domain Expert (Dean of Academics)** – Validation against academic workflows
- **Project Evaluators** – Assessment of requirement depth and system modeling

1.2 Version History

Version	Date	Description	Author
1.00	2026-01-30	Initial draft based on Dean interview	Team 39_endur
1.01	2026-01-31	Refined system requirements	Team 39_endur
1.02	2026-01-31	Added Dean user & system requirements	Team 39_endur
1.03	2026-02-01	Finalized requirements specification	Team 39_endur

2. Introduction

2.1 Need for the System

The current end-semester feedback mechanism fails in two critical areas:

 Pain Point 1 – Timing Latency

Feedback arrives too late to benefit the current batch and is often distorted by *revenge ratings* submitted after grade disclosure.

Pain Point 2 – Engagement Fatigue

Only ~**30% student participation** due to feedback being perceived as a compulsory chore rather than meaningful input.

2.2 System Functions

- **Continuous Feedback Cycles** – Multiple review windows per semester
 - **Role-Based Dashboards** – Student, Faculty, HOD, and Dean views
 - **Gap Analysis** – Automated comparison of self-reflection vs student perception
 - **Anonymity Preservation** – Complete decoupling of student identity
-

3. Glossary

Refer to `definitions.yaml` in the repository for the complete ubiquitous language.

- **FeedbackCycle** – Time-bound feedback collection window
 - **GapAnalysis** – Absolute difference between self-rating and student average
 - **AnonymizedReport** – Identity-stripped aggregated feedback report
 - **ActionReport** – Faculty remediation plan submitted to HOD
 - **ReviewOfReviews** – Meta-evaluation of feedback system effectiveness
 - **FeedbackTrend** – Longitudinal performance score analysis
 - **SelfReflection** – Faculty or student self-assessment
 - **ReviewCheckIn** – Structured faculty–HOD progress discussion
 - **FeedbackResponse** – Immutable atomic student feedback submission
-

4. User Requirements Definition

4.1 Student Services

- **UR-01:** Submit lightweight weekly/monthly feedback (<30 seconds)
- **UR-02:** View history of submitted feedback
- **UR-03:** Participate in monthly ReviewOfReviews feedback

4.2 Faculty Services

- **UR-04:** Submit Self-Reflection before viewing student feedback
- **UR-05:** View personal performance and FeedbackTrends
- **UR-06:** Submit ActionReports when alerts are triggered

4.3 Administrative (HOD) Services

- **UR-07:** View department-level aggregated reports
- **UR-08:** Conduct ReviewCheckIns with faculty
- **UR-09:** Define EvaluationParameters per FeedbackCycle

4.4 Dean Services

- **UR-10:** View institution-wide performance analytics
 - **UR-11:** Resolve conflict-of-interest review cases
 - **UR-12:** Review all ActionReports and ReviewCheckIns
 - **UR-13:** Track long-term institutional FeedbackTrends
 - **UR-14:** Audit compliance and flagged submissions
 - **UR-15:** Freeze, reopen, or invalidate FeedbackCycles with audit logging
-

5. System Requirements Specification

5.1 Functional Requirements

FR-01: FeedbackCycle Window Enforcement

- Reject submissions outside configured timestamps
- Return explicit validation errors
- Uniform enforcement across all cycle types

FR-02: Attendance-Weighted PerformanceScore

- Weight = 1.0 if attendance ≥ 75%
- Thresholds configurable
- Fallback to unweighted averaging if attendance unavailable

FR-03: GapAnalysis Computation

$$\text{Gap} = |\text{SelfReflection.Score} - \text{Avg}(\text{StudentScore})|$$

FR-04: ActionReport Immutability

- Read-only after submission
- No edits or deletions
- New submission only in subsequent cycles

FR-05: AnonymizedReport Generation

- Aggregated parameter scores

- No student identity exposure
- Immutable after generation

FR-06: FeedbackTrend Computation

- Computed per EvaluationParameter
- Historical trends remain unchanged

FR-07: ComplianceAudit

- Detect random ticking and suspicious patterns
- Exclude flagged submissions
- Append-only audit logs

FR-08: ReviewCheckIn Records

- Linked to trends, gaps, and actions
- Immutable once saved

FR-09: Institutional Performance Aggregation

- Cross-department analytics
- Comparative department ranking

FR-10: Conflict Resolution Workflow

- Dean assumes authority when HOD is reviewed
- Self-review prevention enforced

FR-11: Emergency Cycle Management

- Dean-triggered freeze with mandatory justification

FR-12: Long-Term Trend Analysis

- Multi-year aggregation
- Course content analyzed independently
- Historical data immutable

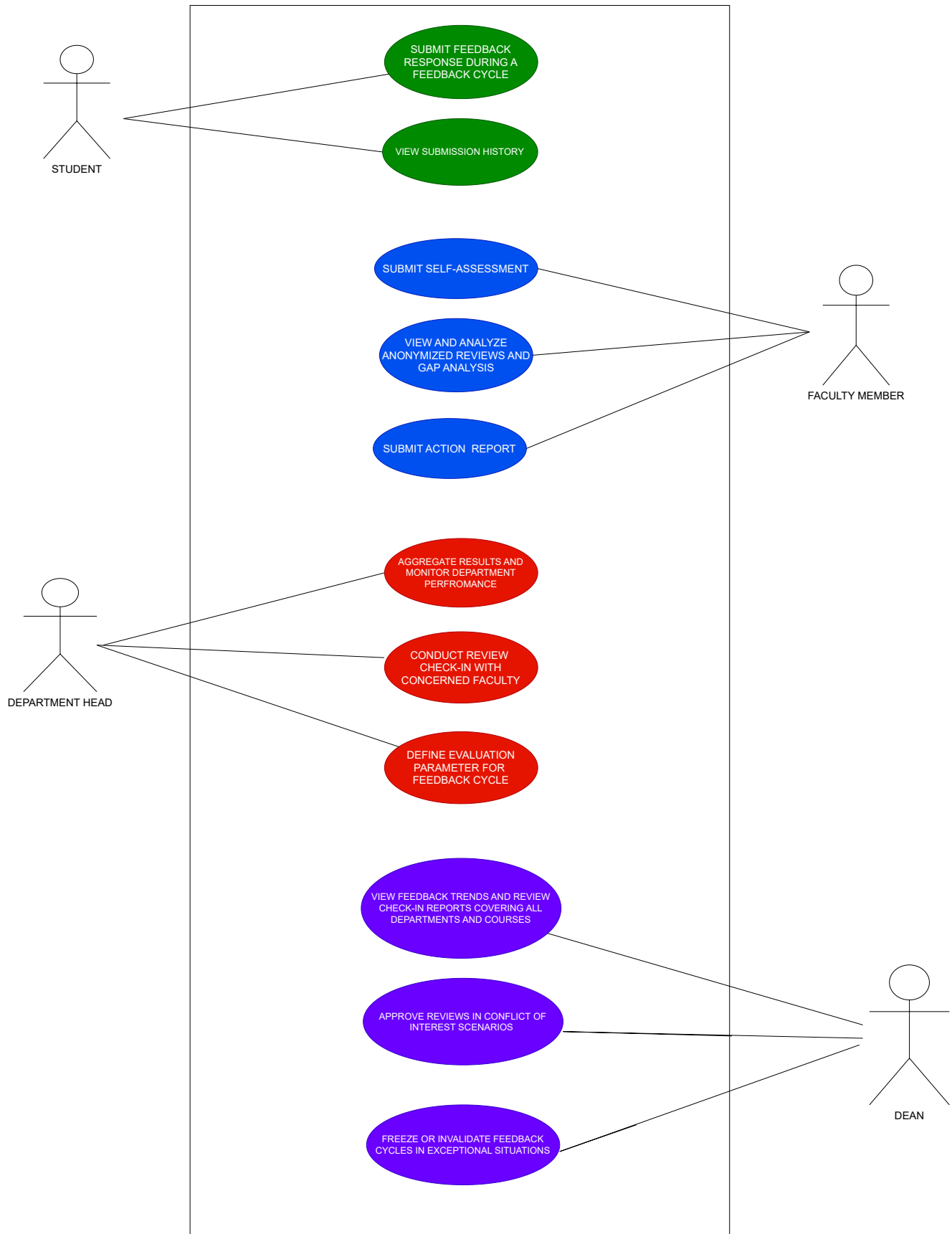
5.2 Non-Functional Requirements

- **NFR-01:** Absolute anonymity and privacy
- **NFR-02:** ≥ 500 concurrent writes in peak windows
- **NFR-03:** 99.9% availability during critical periods

- **NFR-04:** Immutable finalized artifacts
 - **NFR-05:** Append-only audit logging
-

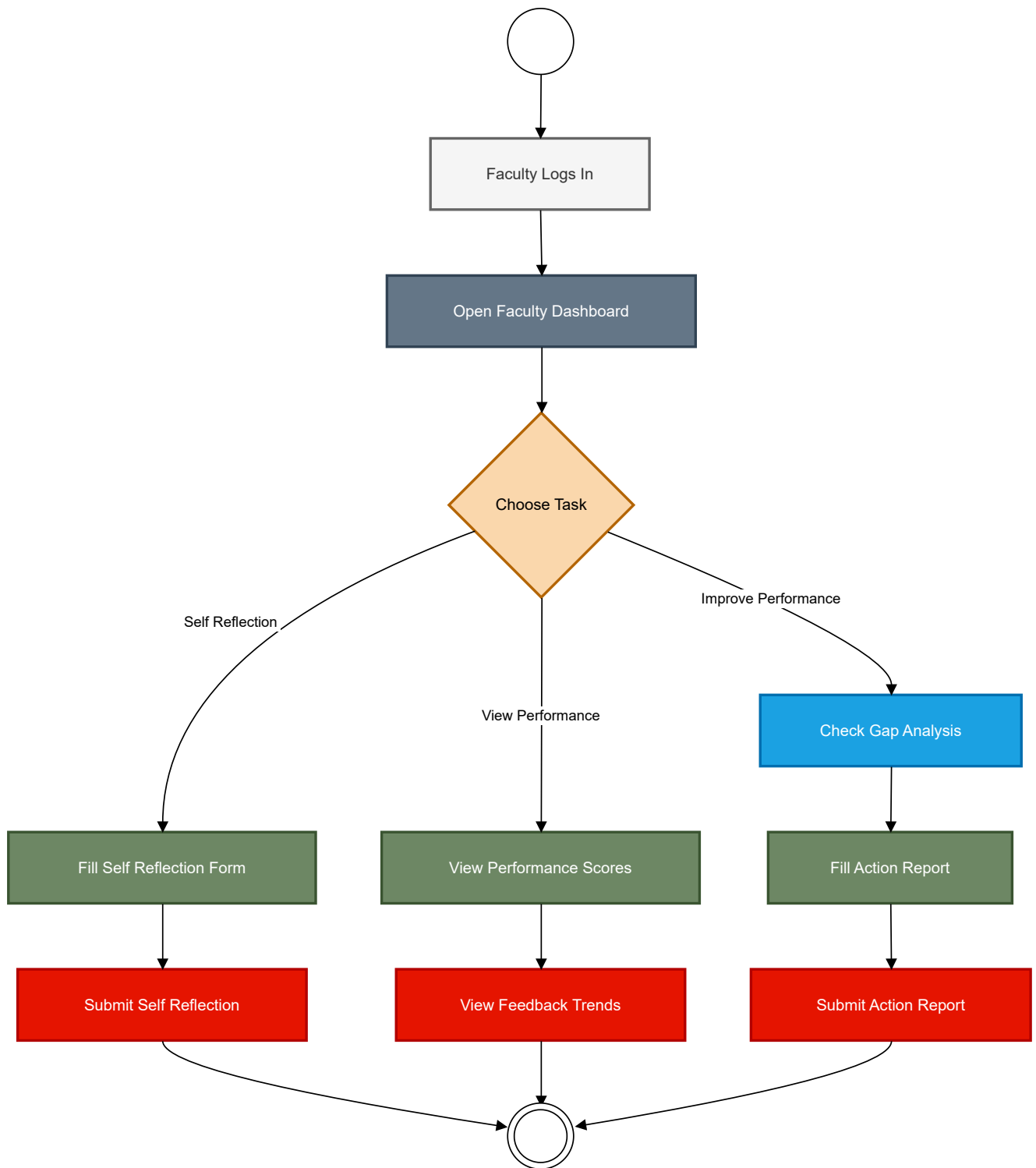
6. System Models

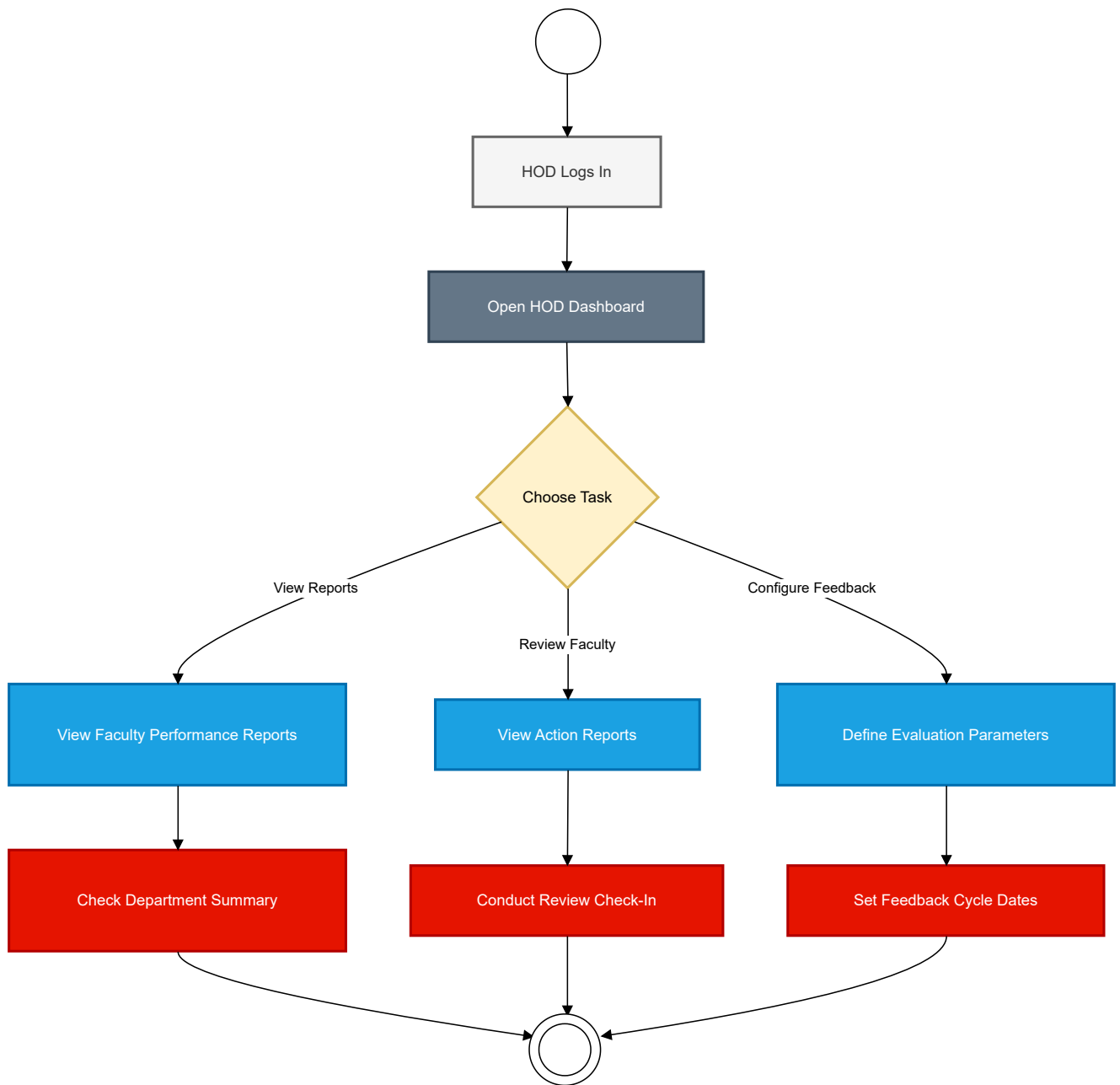
6.1 Use Case Diagram

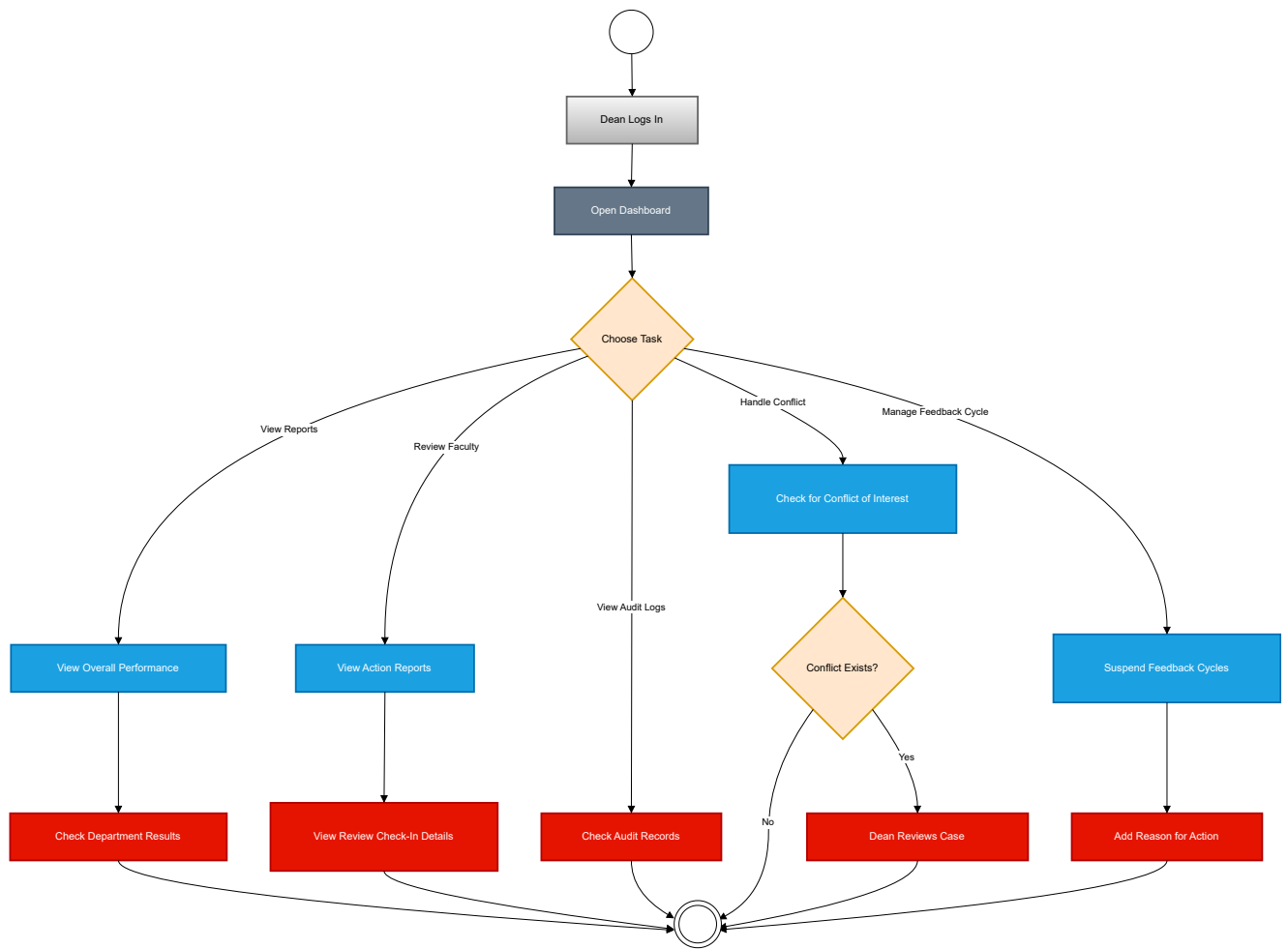


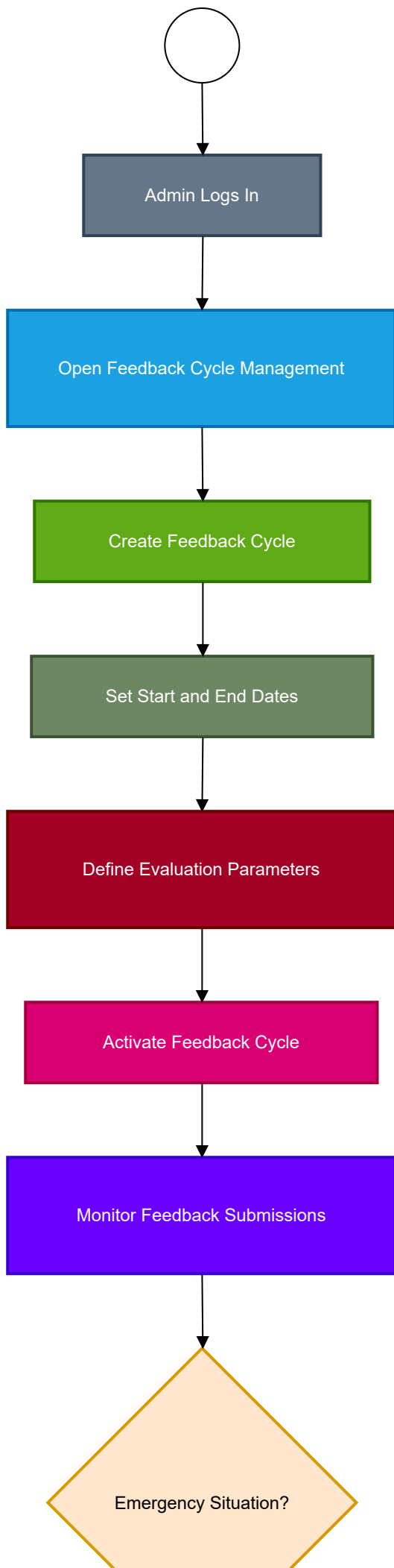
6.2 Activity Diagrams

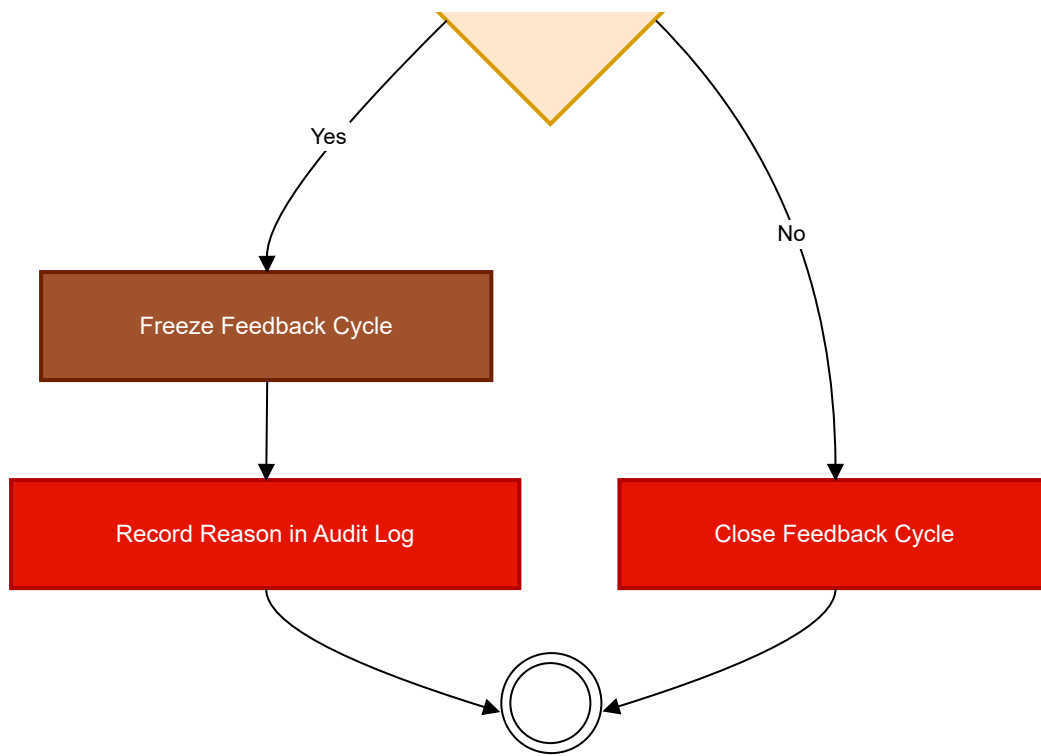




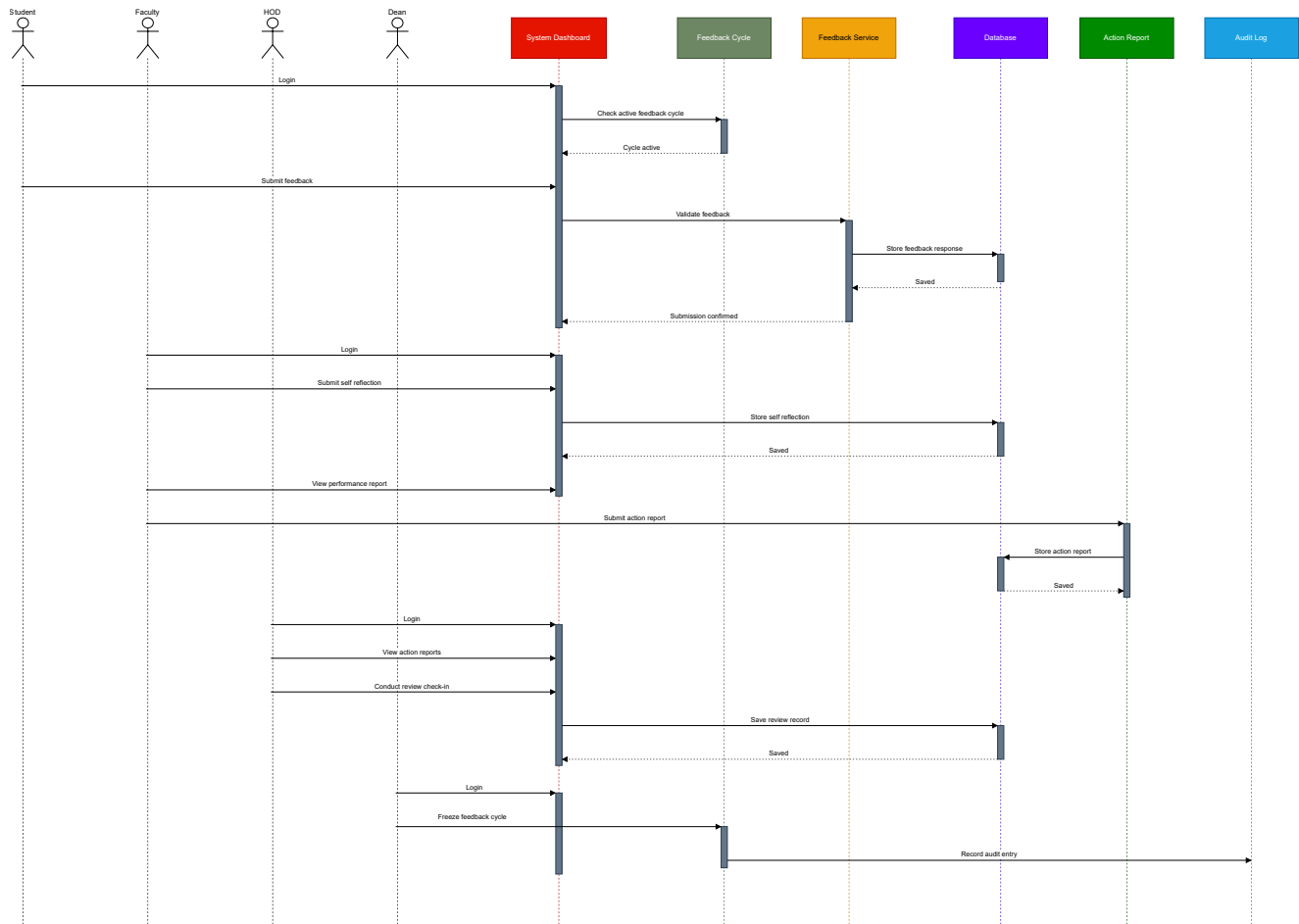








6.3 Sequence Diagram



7. Index

A: ActionReport, AnonymizedReport, AttendanceWeightedFeedback

C: ComplianceAudit, CourseOffering

D: DepartmentHead, Dean

F: FacultyMember, FeedbackCycle, FeedbackResponse, FeedbackTrend

G: GapAnalysis

R: ReviewCheckIn, ReviewOfReviews